

De Novo Protein–Ligand Design Including Protein Flexibility and Conformational Adaptation

Jakob Agamia Martin Zacharias

Technical University Munich

Bioinformatics, 2026, 42(2), bttag027

1 Introduction

2 Materials and Methods

- MC Simulation Approach
- Movesets
- Scoring of Compounds During MC Simulation
- Molecular Dynamics Simulations and MMGBSA Calculations

3 Results and Discussion

- MC-Based Recovery of Target Ligands
- Correlation of Chai-1 Confidence Score and Ligand Binding Affinity
- *De Novo* Generation Using Atomistic-Step MC Simulation

4 Conclusion

Introduction

The problem

Designing drugs means finding small molecules that bind to a specific protein — but proteins are not rigid locks.

- Many drugs are small organic molecules that bind a target protein and alter its function
- The search happens in a virtually infinite chemical space
- Traditional methods assume a fixed binding pocket, ignoring induced fit — a critical limitation

The opportunity

AI structure predictors make protein flexibility available at every evaluation step — this is the paper's core insight.

- AlphaFold successors can predict a protein–ligand complex from sequence + SMILES in seconds
- Explore this vast chemical space with a stochastic search guided by

MC Simulation Approach

AI-MCLig grows a ligand from benzene by applying random chemical operations, accepting or rejecting each change based on how much it improves the Chai-1 score.

- Start from benzene (the simplest aromatic scaffold) as the initial SMILES
- At each step: apply one chemical operation from a **moveset**
- Rebuild the full protein–ligand complex with Chai-1 from scratch → get new pLDDT score
- Accept or reject using the **Metropolis criterion**:

$$P(\Delta s) = \exp(-\beta \cdot \Delta s)$$

9 Atomistic moves $\beta = 50$

Less exploratory — fine-grained edits.

Nine atom-level chemical operations (add/remove atoms and groups, change atom or bond types, form/break rings). Probabilities empirically tuned on three protein targets.

BRICS fragment moves $\beta = 5$

More exploratory — larger jumps.

Instead of atom-level edits, the fragment-based variant recombines whole molecular building blocks. Larger score changes per step require a lower β to maintain a reasonable acceptance rate.

Scoring of Compounds During MC Simulation

The score is not just Chai-1 — three drug-property biases steer the search toward synthesisable, soluble, drug-like molecules.

$$\text{score} = 0.8 \cdot \text{Chai-1} - 0.1 \cdot \text{SA} + 0.05 \cdot \text{ESOL} + 0.05 \cdot \text{QED}$$

- **Chai-1** (0.8): pLDDT confidence of the predicted complex — proxy for binding affinity
- **SA** (−0.1): synthetic accessibility score — penalises hard-to-synthesise molecules; capped at a minimum complexity threshold so the ligand is large enough to fill the pocket
- **ESOL** (+0.05): estimated aqueous solubility (Delaney 2004) — favours water-soluble compounds
- **QED** (+0.05): quantitative estimate of drug-likeness (Bickerton et al. 2012, without MW factor) — favours Lipinski-compliant molecules

Molecular Dynamics Simulations and MMGBSA Calculations

What is MMGBSA?

- Estimates the free energy of binding
- Based on MD simulation + implicit solvation
- More negative = stronger binding
- Typical drug range: -15 to -50 kcal/mol

Why use it here?

- Chai-1 gives a structural confidence score; MMGBSA is classical physics (force field + thermodynamics)
- Two methods with different assumptions that **agree** $\Rightarrow$ stronger evidence of correctness

Result

Generated compounds show MMGBSA values between -19 and -57 kcal/mol — the same range as experimentally known binders (Table 3).

MC-Based Recovery of Target Ligands from Simple Starting Compounds

Does the chemical-space engine work? Replacing Chai-1 with dice similarity to a known ligand, the MC search rediscovers the target compound from benzene in most cases.

- **Validation strategy:** swap the oracle score for **Dice similarity** to a known target compound — if the search works, it should recover the target
- **Result (Table 1):** 8 out of 11 compounds recovered with Dice score = 1.00; the remaining 3 reached Dice ≥ 0.73
- Some complex compounds got stuck in local minima due to the rough chemical-space landscape — not a failure, the search still finds chemically close solutions
- Raising the temperature (lowering β) allows more exploration but reduces selectivity — the two-stage protocol (30 parallel runs $\times$ 5000 steps, then 1 run $\times$ 10 000 steps) is used instead

About These Validations

The two validations are complementary, not redundant:

Validation	What it tests	When
Recovery test (Table 1)	Does the search engine work?	Before real design
MMGBSA + Boltz-2 (Table 2)	Are the generated compounds plausible binders ?	After real design

Chai-1: The Imperfect Oracle

Chai-1 pLDDT is not a direct affinity measurement — but its average trend is positive, and that is enough to guide a Monte Carlo search uphill.

- Three protein targets evaluated: bromodomain, Pim-1 kinase, p38 kinase
- Experimental binding affinities (kcal/mol) from PDBbind (Liu et al. 2017) compared to Chai-1 pLDDT scores
- **Key result:** no direct point-to-point correlation; but the **average trend** across binned Chai-1 scores is consistently positive for all three targets (Figure 1)
- Autodock Vina tested as alternative oracle — no significant improvement in correlation
- **Conclusion:** the oracle is imperfect but good enough to guide the search

De Novo Generation: Targets and Scores

From benzene to a drug-like compound in 2 000 steps: AI-MCLig generates novel ligands whose MMGBSA and Boltz-2 scores match experimentally known binders.

- Applied to four targets: bromodomain, p38 kinase, Pim-1 kinase, β -1 adrenergic receptor (GPCR)
- 2 000 MC steps, $\beta = 50$, starting from benzene; **3 independent runs per target** (~ 20 h each on NVIDIA RTX4090)
- Final Chai-1 scores: **0.85–0.92** across all 12 compounds (Table 2)
- MMGBSA scores: **–19 to –57 kcal/mol** — in the same range as experimentally validated high-affinity binders (Table 3)

De Novo Generation: Convergence and Protein Adaptation

- Score improves rapidly in the **first 250 steps**, then more gradually (Figure 3A)
- Pocket RMSD (Figure 3C):
 - **Bromodomain** stays low (~ 0.5 Å) — rigid pocket
 - **p38 kinase** and **Pim-1 kinase** show larger fluctuations (0.5–2.0 Å) — flexible pockets adapting to each candidate ligand
- **Cleanup phase:** 100 additional steps optimising SA, ESOL, QED with a Chai-1 threshold constraint — improves drug-likeness without losing the binding score
- UMAP analysis: structurally diverse outputs per target all land near known-binder clusters in chemical space

Conclusion

AI-MCLig demonstrates that a simple Monte Carlo search scored by an AI structure predictor is sufficient to generate novel drug-like ligands with predicted binding energies comparable to known experimental binders.

- **Central claim:** Chai-1 as a scoring oracle inside an MC loop enables *de novo* ligand generation with **full protein flexibility** — no rigid pocket assumption
- **Validated on 4 targets:** bromodomain, p38 kinase, Pim-1 kinase, β -1 adrenergic receptor — diverse target classes
- **MMGBSA and Boltz-2 scores** of generated compounds overlap with experimentally known high-affinity binders (Tables 2 & 3)
- **Complementary, not competing:** supplements traditional docking and deep generative models; adaptable to scaffold optimisation, side-chain refinement, or fragment recombination

Ask Us Something

- Why Chai-1 as oracle rather than a formula-based physical energy evaluation?
- Why those specific probabilities for each move?
- What is β ? (Metropolis acceptance criterion)
- Difference between *ligand* and *binder*
- Why start with benzene?
- Is high affinity always the functional requirement?
- Once the score plateaus after ~ 250 moves, the MC engine seems inefficient — how would you fix it?